THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ladder 85 m long reaches 75 m up a wall. Its top slides down 7 m along the wall. The foot of the ladder moves away from the wall by (in m): 1

(a) 7 **(b)** 40 **(c)** 51 **(d)** 11

2. The legs of a right triangle are in the ratio $9 : 40$ and its hypotenuse is 82 cm. Its perimeter (in cm) is: 1

(a) 180 **(b)** 98 **(c)** 164 **(d)** 182

3. The sides of a rectangle are in the ratio $20 : 21$ and its diagonal is 29 cm. The area of the rectangle (in cm²) is: 1

(a) 29 **(b)** 82 **(c)** 421 **(d)** 420

4. A Pythagorean triplet whose smallest member is 26 is: 1

(a) 26, 168, 170 **(b)** 26, 167, 169 **(c)** 26, 27, 28 **(d)** 26, 169, 171

5. A rhombus has perimeter 356 cm and one diagonal 78 cm. Its area (in cm²) is: 1

(a) 3120 **(b)** 6279 **(c)** 6942 **(d)** 6240

6. A ladder 41 m long, standing in a street, rests against a house on one side when its top reaches 40 m. Without moving its foot it is turned to rest against the opposite house, reaching 9 m. The width of the street (in m) is: 1

(a) 51 **(b)** 41 **(c)** 49 **(d)** 50

7. A square has a diagonal of $12\sqrt{2}$ cm. Its area (in cm^2) is: 1

(a) 288 **(b)** 576 **(c)** 144 **(d)** 72

8. A ladder 65 m long reaches 56 m up a wall. Its top slides down 4 m along the wall. The foot of the ladder moves away from the wall by (in m): 1

(a) 33 **(b)** 6 **(c)** 39 **(d)** 4

9. **Assertion (A):** The diagonal of a rectangle of length 12 cm and breadth 5 cm is 13 cm. 1
Reason (R): The diagonal divides the rectangle into two right-angled triangles whose hypotenuse is the diagonal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A triangle with sides 4 cm, 5 cm and 6 cm is not right-angled. 1
Reason (R): In every triangle the sum of the squares of the two shorter sides equals the square of the longest side.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A kite string 85 m long is stretched straight and the kite is 77 m above the boy's hand. Find the horizontal distance of the boy from the point directly below the kite. 2

12. The legs of a right triangle are in the ratio $7 : 24$ and the hypotenuse is 75 cm. Find the legs. 2

13. Find a Pythagorean triplet in which one member is 36. 2

14. A kite string 89 m long is stretched straight and the kite is 80 m above the boy's hand. Find the horizontal distance of the boy from the point directly below the kite. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 85 m long, with its foot in a street, reaches a window 84 m high on one side. Without moving its foot it is turned to reach a window 40 m high on the opposite side. Find the width of the street. 3

16. A rhombus has perimeter 388 cm and one diagonal 130 cm. Find the other diagonal and the area. 3

17. Two ships leave a port, one sailing east at 16 km/h and the other south at 30 km/h. How far apart are they after 5 hours? 3

OR

A ladder 73 m long, with its foot in a street, reaches a window 55 m high on one side. Without moving its foot it is turned to reach a window 48 m high on the opposite side. Find the width of the street. 3

18. A rhombus has perimeter 436 cm and one diagonal 120 cm. Find the other diagonal and the area. 3

OR

Two ships leave a port, one sailing east at 24 km/h and the other south at 7 km/h. How far apart are they after 3 hours? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A carpenter checks whether the corners of triangular frames are right angles using side lengths. Frame P has sides 9, 12, 15 cm; Frame Q has sides 7, 24, 25 cm; Frame R has sides 8, 10, 12 cm. 5
(i) Is frame P right-angled?
(ii) Is frame Q right-angled?
(iii) Is frame R right-angled?
(iv) A fourth frame is right-angled with perpendicular sides 20 cm and 21 cm; find its hypotenuse and perimeter.

20. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 7 cm, BC = 24 cm, CD = 60 cm and DA = 65 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A ladder 25 m long reaches a point 24 m up a wall. Its top slides down 4 m. Find 5
(i) the original distance of the foot from the wall,
(ii) its new distance from the wall,
(iii) how far the foot moves.

END OF QUESTION PAPER — Total: 40 Marks